In Silico Design and Preliminary Evaluation of CRISPR-Cas9 Guide RNAs Targeting the HBB Gene for Sickle Cell Disease Correction

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Abstract:

Sickle cell disease (SCD) is a genetic disorder caused by a point mutation (GAG to GTG) in the HBB gene. This mutation replaces glutamic acid with valine (Glu6Val). This study offers a computational method to design and assess CRISPR-Cas9 guide RNAs targeting the mutation site. The HBB mRNA reference sequence (NM_000518.5) was retrieved and used for preliminary target identification, and candidate guide RNAs were generated using the CHOPCHOP CRISPR design tool. Among the identified guides, the sequence TCTGCCGTTTACTGCCCTGTGG showed the highest predicted efficiency at 66.74% and an optimal GC content of 60%. Off-target analysis revealed moderate risks associated with mismatches. We suggest using a homology-directed repair (HDR) strategy for correcting the mutation. We also discuss advanced genome editing methods like base editing and prime editing to improve accuracy. This study shows the potential of computational genome engineering in creating targeted treatment strategies.

However, as CRISPR-Cas9 targets genomic DNA, further validation using genomic sequences is required to ensure accurate targeting.

Introduction:

Sickle cell disease (SCD) is a genetic blood disorder caused by a single nucleotide change in the β-globin gene (HBB). In this case, the codon GAG is changed to GTG. This swap replaces glutamic acid with valine, resulting in abnormal haemoglobin and serious clinical issues.

CRISPR-Cas9 is a powerful tool for editing genomes. It can make targeted changes in DNA. The system uses a guide RNA (gRNA) to direct the Cas9 nuclease to a specific spot in the genome next to a protospacer adjacent motif (PAM), usually NGG. When it binds, Cas9 creates a double-strand break. This break is repaired by cellular mechanisms like non-homologous end joining (NHEJ) or homology-directed repair (HDR).

The computational design of CRISPR guide RNAs is an important first step before validation through experiments. It allows for quick screening of potential guides, assessing their efficiency, and predicting off-target effects. This approach saves time and money in lab studies.

Recent advances, including approved CRISPR therapies for sickle cell disease, show the potential of genome editing in clinical settings. In this context, the current study aims to design and evaluate candidate guide RNAs that target the HBB mutation site using computational tools and to address important methodological issues.

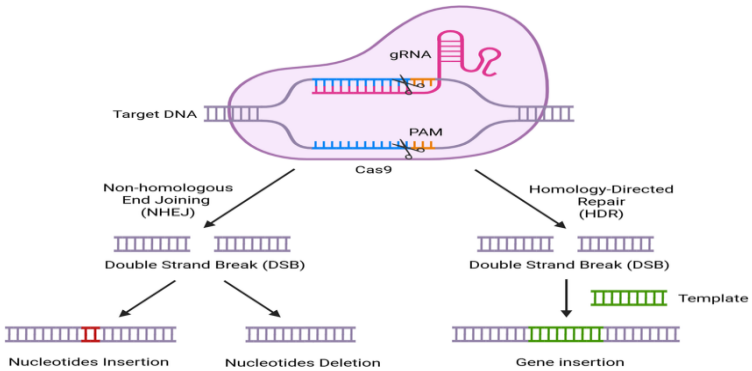

Figure: CRISPR-Cas9 mechanism showing gRNA targeting, PAM recognition, DNA cleavage, and repair via NHEJ or HDR.

Materials and Methods:

Sequence Retrieval

We obtained the human HBB mRNA sequence (NM_000518.5) from the NCBI database. This sequence was chosen because it represents the coding region without introns, making it suitable for guide RNA design.

Target Region Selection

We selected a sequence around the sickle cell mutation site (Glu6Val) for analysis. The mutation involves changing the codon from GAG (wild type) to GTG (mutant), making it a suitable target for CRISPR-mediated correction.

Guide RNA Design

We used the CHOPCHOP CRISPR design tool to create guide RNAs with these parameters:

- Organism: Homo sapiens (hg38)
- Editing system: CRISPR/Cas9
- PAM sequence: NGG

An expanded sequence region ensured we included PAM sites needed for guide RNA binding.

Figure: Input parameters used in the CHOPCHOP tool showing target sequence selection, organism (Homo sapiens), and CRISPR/Cas9 system configuration.

Evaluation Criteria

We evaluated the generated guide RNAs based on:

- GC content (optimal range: 40 to 60%)
- Predicted efficiency score
- Off-target potential (measured using mismatch scores MM0 to MM3)

Methodological Limitation and Correction

CRISPR-Cas9 targets genomic DNA within the nucleus, whereas this study utilized mRNA sequence data for initial guide RNA design. As mRNA sequences represent spliced transcripts, there is a possibility that designed guide RNAs may span exon-exon junctions, which are not contiguous in genomic DNA. Therefore, the guide RNAs identified in this study should be considered preliminary and require validation using genomic DNA sequences of the HBB gene to ensure accurate targeting.

Results:

We identified multiple guide RNA candidates within the chosen HBB target region.

Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	GTAACGGCAGACTTCTCCTCAGG	seq:19	-	55	2	1	0	0	4	51.76
2	GTCTGCCGTTACTGCCCTGTGGG	seq:30	+	60	1	1	0	1	4	46.54
3	CACGTTCACCTTGCCCCACAGGG	seq:44	-	60	0	1	1	0	5	0.00
4	CGTTACTGCCCTGTGGGGCAAGG	seq:36	+	65	2	1	0	1	6	46.69
5	AGTCTGCCGTTACTGCCCTGTGG	seq:29	+	60	1	1	0	1	7	65.81
6	CTTGCCCCACAGGGCAGTAACGG	seq:35	-	60	2	1	0	0	15	48.52
7	TCTGCCGTTACTGCCCTGTGGGG	seq:31	+	60	1	1	0	1	17	66.74
8	CCTGTGGGGCAAGGTGAACGTGG	seq:45	+	65	0	1	3	1	13	0.00
9	CATGGTGCATCTGACTCCTGAGG	seq:3	+	55	0	2	0	3	14	0.00
10	CTCAGGAGTCAGATGCACCATGG	seq:2	-	55	0	2	0	1	34	0.00

Figure: Predicted guide RNA candidates generated using CHOPCHOP showing sequence, PAM sites, GC content, and efficiency scores.

The top-performing guide RNA sequence was:

TCTGCCGTTTACTGCCCTGTGG

This guide had the highest predicted efficiency of 66.74% and a GC content of 60%, indicating strong binding stability and cleavage capability.

A secondary candidate guide RNA was:

AGTCTGCCGTTTACTGCCCTGTGG

This guide had a similar efficiency score of 65.81% and comparable GC content, making it a good alternative.

Our analysis revealed that having PAM sequences (NGG) near the mutation site was crucial for effective guide RNA design. Off-target analysis showed one perfect match (MM0 = 1) and moderate risks from mismatched off-target sites, pointing to the need for further optimisation.

Guide Sequence	Spacer (20nt)	PAM	Efficiency (%)	GC (%)	MM0	MM3
TCTGCCGTTTACTGCCCTGTGGG	TCTGCCGTTTACTGCCCTGT	GGG	66.74	60	1	17
AGTCTGCCGTTTACTGCCCTGTGG	AGTCTGCCGTTTACTGCCCT	GGG	65.81	60	1	7

Discussion:

The results show that computational tools can effectively identify candidate CRISPR guide RNAs that target the HBB mutation region. The selected guides have favorable efficiency and GC content.

However, off-target effects are still a major concern in genome editing. To improve specificity, we can use:

- High-fidelity Cas9 variants
- Truncated guide RNAs
- Chemical modifications
- Mutation Correction Strategy

A homology-directed repair (HDR) approach is proposed:

- Introduce a donor DNA template with the correct GAG codon
- Cas9 induces a double-strand break near the mutation
- Cellular repair restores the wild-type sequence

Improved Editing Approaches

To enhance precision:

- Base editing allows direct nucleotide conversion without double-strand breaks
- Prime editing enables highly precise sequence modification

These technologies provide promising alternatives for safer therapeutic applications.

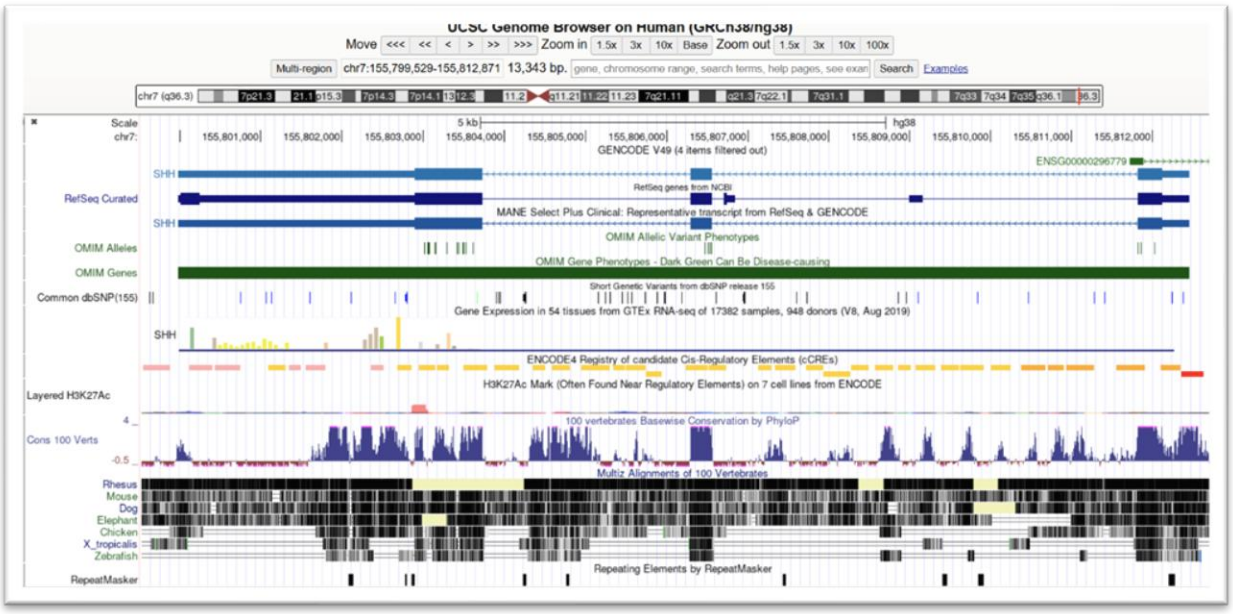

Figure: Genomic organization of the HBB gene showing exon–intron structure, highlighting the importance of genomic DNA-based CRISPR targeting.

Future work should include validation using genomic DNA sequences and in vitro experimental verification to confirm guide RNA efficiency and specificity.

Conclusion:

This study demonstrates the feasibility of designing CRISPR-Cas9 guide RNAs targeting the HBB gene mutation using an in silico approach. While the identified guides show promising efficiency, the use of mRNA sequences introduces limitations that require correction through genomic validation. Future work should focus on designing guide RNAs using genomic DNA and experimentally validating their efficiency and specificity.

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